

Solved Examples of Conditional Probability, Law of Total Probability, and Bayes' Theorem

Example 1: A bag contains 5 red balls and 3 green balls. If a ball is drawn at random and it is red, what is the probability that a second ball drawn (without replacement) is also red?

Solution:

- Total balls initially: $5 + 3 = 8$.
- After drawing 1 red ball, the bag has 4 red and 3 green balls.
- Probability of drawing a second red ball:

$$P = \frac{4}{7}.$$

Example 2: In a class, 60% of students are girls, and 40% of girls wear glasses. What is the probability that a randomly selected student is a girl who wears glasses?

Solution:

- Let $P(G) = 0.6$ (probability of being a girl).
- Let $P(W|G) = 0.4$ (probability of wearing glasses given the student is a girl).
- Probability of being a girl who wears glasses:

$$P(G \cap W) = P(G) \cdot P(W|G) = 0.6 \cdot 0.4 = 0.24.$$

Example 3: A fair die is rolled. What is the probability that the number is even, given that it is greater than 3?

Solution:

- Sample space for numbers greater than 3: $\{4, 5, 6\}$.
- Even numbers in this set: $\{4, 6\}$.
- Probability:

$$P = \frac{2}{3}.$$

Example 4: In a deck of 52 cards, what is the probability of drawing a heart, given that the card drawn is a queen?

Solution:

- There are 4 queens in a deck, and 1 of them is a heart.
- Probability:

$$P = \frac{1}{4}.$$

Example 5: A box contains 3 blue marbles and 2 red marbles. If two marbles are drawn without replacement, what is the probability that the second marble is red, given that the first marble is blue?

Solution:

- After drawing 1 blue marble, the box has 2 blue and 2 red marbles.
- Probability of drawing a red marble:

$$P = \frac{2}{4} = \frac{1}{2}.$$

Example 6: A factory has two machines, A and B. Machine A produces 60% of the items, and Machine B produces 40%. The defect rate for Machine A is 2%, and for Machine B is 3%. What is the probability that a randomly selected item is defective?

Solution:

- Let $P(A) = 0.6$, $P(B) = 0.4$.
- Let $P(D|A) = 0.02$, $P(D|B) = 0.03$.
- Total probability of defect:

$$P(D) = P(A) \cdot P(D|A) + P(B) \cdot P(D|B) = 0.6 \cdot 0.02 + 0.4 \cdot 0.03 = 0.024.$$

Example 7: In a school, 70% of students are in Class X, and 30% are in Class Y. The pass rate for Class X is 80%, and for Class Y is 90%. What is the probability that a randomly selected student passes?

Solution:

- Let $P(X) = 0.7$, $P(Y) = 0.3$.
- Let $P(P|X) = 0.8$, $P(P|Y) = 0.9$.
- Total probability of passing:

$$P(P) = P(X) \cdot P(P|X) + P(Y) \cdot P(P|Y) = 0.7 \cdot 0.8 + 0.3 \cdot 0.9 = 0.83.$$

Example 8: A bag contains 4 red balls and 6 blue balls. Two balls are drawn with replacement. What is the probability that the second ball is red?

Solution:

- The probability of drawing a red ball is the same for each draw:

$$P(R) = \frac{4}{10} = 0.4.$$

Example 9: A company has two departments, A and B. Department A employs 60% of the workforce, and Department B employs 40%. The promotion rate in Department A is 10%, and in Department B is 15%. What is the probability that a randomly selected employee gets promoted?

Solution:

- Let $P(A) = 0.6$, $P(B) = 0.4$.
- Let $P(P|A) = 0.1$, $P(P|B) = 0.15$.
- Total probability of promotion:

$$P(P) = P(A) \cdot P(P|A) + P(B) \cdot P(P|B) = 0.6 \cdot 0.1 + 0.4 \cdot 0.15 = 0.12.$$

Example 10: A box contains 3 white balls and 2 black balls. Two balls are drawn without replacement. What is the probability that the second ball is white?

Solution:

- Use the law of total probability:
 - If the first ball is white: $P(W_2|W_1) = \frac{2}{4} = 0.5$.
 - If the first ball is black: $P(W_2|B_1) = \frac{3}{4} = 0.75$.
- Total probability:

$$P(W_2) = P(W_1) \cdot P(W_2|W_1) + P(B_1) \cdot P(W_2|B_1) = \frac{3}{5} \cdot 0.5 + \frac{2}{5} \cdot 0.75 = 0.6.$$

Example 11: A disease affects 1% of the population. A test for the disease is 99% accurate (true positive rate) but has a 2% false positive rate. If a person tests positive, what is the probability they have the disease?

Solution:

- Let $P(D) = 0.01$, $P(\neg D) = 0.99$.
- Let $P(T|D) = 0.99$, $P(T|\neg D) = 0.02$.
- Using Bayes' Theorem:

$$P(D|T) = \frac{P(T|D) \cdot P(D)}{P(T|D) \cdot P(D) + P(T|\neg D) \cdot P(\neg D)} = \frac{0.99 \cdot 0.01}{0.99 \cdot 0.01 + 0.02 \cdot 0.99} = 0.333.$$

Example 12: A company has two machines, A and B. Machine A produces 60% of the items, and Machine B produces 40%. The defect rate for Machine A is 2%, and for Machine B is 3%. If an item is defective, what is the probability it was produced by Machine A?

Solution:

- Let $P(A) = 0.6$, $P(B) = 0.4$.
- Let $P(D|A) = 0.02$, $P(D|B) = 0.03$.
- Using Bayes' Theorem:

$$P(A|D) = \frac{P(D|A) \cdot P(A)}{P(D|A) \cdot P(A) + P(D|B) \cdot P(B)} = \frac{0.02 \cdot 0.6}{0.02 \cdot 0.6 + 0.03 \cdot 0.4} = 0.5.$$

Example 13: A box contains 3 white balls and 2 black balls. Two balls are drawn without replacement. If the second ball is white, what is the probability that the first ball was black?

Solution:

- Let $P(B_1) = \frac{2}{5}$, $P(W_1) = \frac{3}{5}$.
- Let $P(W_2|B_1) = \frac{3}{4}$, $P(W_2|W_1) = \frac{2}{4} = 0.5$.
- Using Bayes' Theorem:

$$P(B_1|W_2) = \frac{P(W_2|B_1) \cdot P(B_1)}{P(W_2|B_1) \cdot P(B_1) + P(W_2|W_1) \cdot P(W_1)} = \frac{\frac{3}{4} \cdot \frac{2}{5}}{\frac{3}{4} \cdot \frac{2}{5} + \frac{1}{2} \cdot \frac{3}{5}} = 0.5.$$

Example 14: A test for a disease is 95% accurate. The disease occurs in 1% of the population. If a person tests positive, what is the probability they have the disease?

Solution:

- Let $P(D) = 0.01$, $P(\neg D) = 0.99$.
- Let $P(T|D) = 0.95$, $P(T|\neg D) = 0.05$.
- Using Bayes' Theorem:

$$P(D|T) = \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + 0.05 \cdot 0.99} = 0.161.$$

Example 15: A box contains 4 red balls and 6 blue balls. Two balls are drawn without replacement. If the second ball is red, what is the probability that the first ball was blue?

Solution:

- Let $P(B_1) = \frac{6}{10} = 0.6$, $P(R_1) = \frac{4}{10} = 0.4$.
- Let $P(R_2|B_1) = \frac{4}{9}$, $P(R_2|R_1) = \frac{3}{9} = \frac{1}{3}$.

- Using Bayes' Theorem:

$$P(B_1|R_2) = \frac{P(R_2|B_1) \cdot P(B_1)}{P(R_2|B_1) \cdot P(B_1) + P(R_2|R_1) \cdot P(R_1)} = \frac{\frac{4}{9} \cdot 0.6}{\frac{4}{9} \cdot 0.6 + \frac{1}{3} \cdot 0.4} = 0.666.$$

Example 16: A box contains 5 red balls and 5 green balls. Two balls are drawn without replacement. What is the probability that both balls are red?

Solution:

- Probability of first ball being red: $P(R_1) = \frac{5}{10} = 0.5$.
- Probability of second ball being red given the first was red: $P(R_2|R_1) = \frac{4}{9}$.
- Total probability:

$$P(R_1 \cap R_2) = P(R_1) \cdot P(R_2|R_1) = 0.5 \cdot \frac{4}{9} = \frac{2}{9}.$$

Example 17: A bag contains 3 red balls and 2 blue balls. Two balls are drawn with replacement. What is the probability that both balls are red?

Solution:

- Probability of drawing a red ball each time: $P(R) = \frac{3}{5}$.
- Since draws are independent, total probability:

$$P(R_1 \cap R_2) = P(R) \cdot P(R) = \frac{3}{5} \cdot \frac{3}{5} = \frac{9}{25}.$$

Example 18: A box contains 4 white balls and 6 black balls. Two balls are drawn without replacement. What is the probability that the balls are of different colors?

Solution:

- Probability of first white and second black:

$$P(W_1 \cap B_2) = \frac{4}{10} \cdot \frac{6}{9} = \frac{24}{90}.$$

- Probability of first black and second white:

$$P(B_1 \cap W_2) = \frac{6}{10} \cdot \frac{4}{9} = \frac{24}{90}.$$

- Total probability:

$$P = \frac{24}{90} + \frac{24}{90} = \frac{48}{90} = \frac{8}{15}.$$

Example 19: A box contains 2 red balls and 3 blue balls. Two balls are drawn without replacement. What is the probability that at least one ball is red?

Solution:

- Probability of no red balls (both blue):

$$P(B_1 \cap B_2) = \frac{3}{5} \cdot \frac{2}{4} = \frac{6}{20} = \frac{3}{10}.$$

- Probability of at least one red ball:

$$P = 1 - P(B_1 \cap B_2) = 1 - \frac{3}{10} = \frac{7}{10}.$$

Example 20: A box contains 5 red balls and 5 green balls. Two balls are drawn without replacement. What is the probability that the balls are of the same color?

Solution:

- Probability of both red:

$$P(R_1 \cap R_2) = \frac{5}{10} \cdot \frac{4}{9} = \frac{20}{90}.$$

- Probability of both green:

$$P(G_1 \cap G_2) = \frac{5}{10} \cdot \frac{4}{9} = \frac{20}{90}.$$

- Total probability:

$$P = \frac{20}{90} + \frac{20}{90} = \frac{40}{90} = \frac{4}{9}.$$

Example 21: A bag contains 6 red balls and 4 green balls. If two balls are drawn without replacement, what is the probability that both balls are red?

Solution:

- Probability of first ball being red: $P(R_1) = \frac{6}{10} = 0.6$.
- Probability of second ball being red given the first was red: $P(R_2|R_1) = \frac{5}{9}$.
- Total probability:

$$P(R_1 \cap R_2) = P(R_1) \cdot P(R_2|R_1) = 0.6 \cdot \frac{5}{9} = \frac{1}{3}.$$

Example 22: A box contains 3 white balls and 5 black balls. If two balls are drawn without replacement, what is the probability that the second ball is black, given that the first ball is white?

Solution:

- After drawing 1 white ball, the box has 2 white and 5 black balls.
- Probability of drawing a black ball:

$$P = \frac{5}{7}.$$

Example 23: A deck of 52 cards is shuffled. What is the probability that the second card drawn is a king, given that the first card drawn is a queen?

Solution:

- After drawing 1 queen, there are 51 cards left, including 4 kings.
- Probability of drawing a king:

$$P = \frac{4}{51}.$$

Example 24: A bag contains 4 red balls and 6 blue balls. If two balls are drawn with replacement, what is the probability that both balls are blue?

Solution:

- Probability of drawing a blue ball each time: $P(B) = \frac{6}{10} = 0.6$.
- Since draws are independent, total probability:

$$P(B_1 \cap B_2) = P(B) \cdot P(B) = 0.6 \cdot 0.6 = 0.36.$$

Example 25: A box contains 2 red balls and 3 green balls. If two balls are drawn without replacement, what is the probability that the second ball is green, given that the first ball is red?

Solution:

- After drawing 1 red ball, the box has 1 red and 3 green balls.
- Probability of drawing a green ball:

$$P = \frac{3}{4}.$$

Example 26: A factory has two machines, A and B. Machine A produces 70% of the items, and Machine B produces 30%. The defect rate for Machine A is 3%, and for Machine B is 5%. What is the probability that a randomly selected item is defective?

Solution:

- Let $P(A) = 0.7$, $P(B) = 0.3$.
- Let $P(D|A) = 0.03$, $P(D|B) = 0.05$.
- Total probability of defect:

$$P(D) = P(A) \cdot P(D|A) + P(B) \cdot P(D|B) = 0.7 \cdot 0.03 + 0.3 \cdot 0.05 = 0.036.$$

Example 27: In a school, 60% of students are in Class X, and 40% are in Class Y. The pass rate for Class X is 85%, and for Class Y is 90%. What is the probability that a randomly selected student passes?

Solution:

- Let $P(X) = 0.6$, $P(Y) = 0.4$.
- Let $P(P|X) = 0.85$, $P(P|Y) = 0.90$.
- Total probability of passing:

$$P(P) = P(X) \cdot P(P|X) + P(Y) \cdot P(P|Y) = 0.6 \cdot 0.85 + 0.4 \cdot 0.90 = 0.87.$$

Example 28: A bag contains 5 red balls and 5 blue balls. Two balls are drawn with replacement. What is the probability that the second ball is red?

Solution:

- The probability of drawing a red ball is the same for each draw:

$$P(R) = \frac{5}{10} = 0.5.$$

Example 29: A company has two departments, A and B. Department A employs 50% of the workforce, and Department B employs 50%. The promotion rate in Department A is 10%, and in Department B is 20%. What is the probability that a randomly selected employee gets promoted?

Solution:

- Let $P(A) = 0.5$, $P(B) = 0.5$.
- Let $P(P|A) = 0.1$, $P(P|B) = 0.2$.
- Total probability of promotion:

$$P(P) = P(A) \cdot P(P|A) + P(B) \cdot P(P|B) = 0.5 \cdot 0.1 + 0.5 \cdot 0.2 = 0.15.$$

Example 30: A box contains 4 white balls and 6 black balls. Two balls are drawn without replacement. What is the probability that the second ball is white?

Solution:

- Use the law of total probability:
 - If the first ball is white: $P(W_2|W_1) = \frac{3}{9} = \frac{1}{3}$.

– If the first ball is black: $P(W_2|B_1) = \frac{4}{9}$.

- Total probability:

$$P(W_2) = P(W_1) \cdot P(W_2|W_1) + P(B_1) \cdot P(W_2|B_1) = \frac{4}{10} \cdot \frac{1}{3} + \frac{6}{10} \cdot \frac{4}{9} = \frac{2}{5}.$$

Example 31: A disease affects 2% of the population. A test for the disease is 95% accurate (true positive rate) but has a 3% false positive rate. If a person tests positive, what is the probability they have the disease?

Solution:

- Let $P(D) = 0.02$, $P(\neg D) = 0.98$.
- Let $P(T|D) = 0.95$, $P(T|\neg D) = 0.03$.
- Using Bayes' Theorem:

$$P(D|T) = \frac{P(T|D) \cdot P(D)}{P(T|D) \cdot P(D) + P(T|\neg D) \cdot P(\neg D)} = \frac{0.95 \cdot 0.02}{0.95 \cdot 0.02 + 0.03 \cdot 0.98} = 0.393.$$

Example 32: A company has two machines, A and B. Machine A produces 60% of the items, and Machine B produces 40%. The defect rate for Machine A is 2%, and for Machine B is 4%. If an item is defective, what is the probability it was produced by Machine A?

Solution:

- Let $P(A) = 0.6$, $P(B) = 0.4$.
- Let $P(D|A) = 0.02$, $P(D|B) = 0.04$.
- Using Bayes' Theorem:

$$P(A|D) = \frac{P(D|A) \cdot P(A)}{P(D|A) \cdot P(A) + P(D|B) \cdot P(B)} = \frac{0.02 \cdot 0.6}{0.02 \cdot 0.6 + 0.04 \cdot 0.4} = 0.428.$$

Example 33: A box contains 3 white balls and 2 black balls. Two balls are drawn without replacement. If the second ball is white, what is the probability that the first ball was black?

Solution:

- Let $P(B_1) = \frac{2}{5}$, $P(W_1) = \frac{3}{5}$.
- Let $P(W_2|B_1) = \frac{3}{4}$, $P(W_2|W_1) = \frac{2}{4} = 0.5$.
- Using Bayes' Theorem:

$$P(B_1|W_2) = \frac{P(W_2|B_1) \cdot P(B_1)}{P(W_2|B_1) \cdot P(B_1) + P(W_2|W_1) \cdot P(W_1)} = \frac{\frac{3}{4} \cdot \frac{2}{5}}{\frac{3}{4} \cdot \frac{2}{5} + \frac{1}{2} \cdot \frac{3}{5}} = 0.5.$$

Example 34: A test for a disease is 90% accurate. The disease occurs in 1% of the population. If a person tests positive, what is the probability they have the disease?

Solution:

- Let $P(D) = 0.01$, $P(\neg D) = 0.99$.
- Let $P(T|D) = 0.90$, $P(T|\neg D) = 0.10$.
- Using Bayes' Theorem:

$$P(D|T) = \frac{0.90 \cdot 0.01}{0.90 \cdot 0.01 + 0.10 \cdot 0.99} = 0.083.$$

Example 35: A box contains 4 red balls and 6 blue balls. Two balls are drawn without replacement. If the second ball is red, what is the probability that the first ball was blue?

Solution:

- Let $P(B_1) = \frac{6}{10} = 0.6$, $P(R_1) = \frac{4}{10} = 0.4$.
- Let $P(R_2|B_1) = \frac{4}{9}$, $P(R_2|R_1) = \frac{3}{9} = \frac{1}{3}$.
- Using Bayes' Theorem:

$$P(B_1|R_2) = \frac{P(R_2|B_1) \cdot P(B_1)}{P(R_2|B_1) \cdot P(B_1) + P(R_2|R_1) \cdot P(R_1)} = \frac{\frac{4}{9} \cdot 0.6}{\frac{4}{9} \cdot 0.6 + \frac{1}{3} \cdot 0.4} = 0.666.$$

Example 36: A box contains 5 red balls and 5 green balls. Two balls are drawn without replacement. What is the probability that both balls are red?

Solution:

- Probability of first ball being red: $P(R_1) = \frac{5}{10} = 0.5$.
- Probability of second ball being red given the first was red: $P(R_2|R_1) = \frac{4}{9}$.
- Total probability:

$$P(R_1 \cap R_2) = P(R_1) \cdot P(R_2|R_1) = 0.5 \cdot \frac{4}{9} = \frac{2}{9}.$$

Example 37: A bag contains 3 red balls and 2 blue balls. Two balls are drawn with replacement. What is the probability that both balls are red?

Solution:

- Probability of drawing a red ball each time: $P(R) = \frac{3}{5}$.
- Since draws are independent, total probability:

$$P(R_1 \cap R_2) = P(R) \cdot P(R) = \frac{3}{5} \cdot \frac{3}{5} = \frac{9}{25}.$$

Example 38: A box contains 4 white balls and 6 black balls. Two balls are drawn without replacement. What is the probability that the balls are of different colors?

Solution:

- Probability of first white and second black:

$$P(W_1 \cap B_2) = \frac{4}{10} \cdot \frac{6}{9} = \frac{24}{90}.$$

- Probability of first black and second white:

$$P(B_1 \cap W_2) = \frac{6}{10} \cdot \frac{4}{9} = \frac{24}{90}.$$

- Total probability:

$$P = \frac{24}{90} + \frac{24}{90} = \frac{48}{90} = \frac{8}{15}.$$

Example 39: A box contains 2 red balls and 3 blue balls. Two balls are drawn without replacement. What is the probability that at least one ball is red?

Solution:

- Probability of no red balls (both blue):

$$P(B_1 \cap B_2) = \frac{3}{5} \cdot \frac{2}{4} = \frac{6}{20} = \frac{3}{10}.$$

- Probability of at least one red ball:

$$P = 1 - P(B_1 \cap B_2) = 1 - \frac{3}{10} = \frac{7}{10}.$$

Example 40: A box contains 5 red balls and 5 green balls. Two balls are drawn without replacement. What is the probability that the balls are of the same color?

Solution:

- Probability of both red:

$$P(R_1 \cap R_2) = \frac{5}{10} \cdot \frac{4}{9} = \frac{20}{90}.$$

- Probability of both green:

$$P(G_1 \cap G_2) = \frac{5}{10} \cdot \frac{4}{9} = \frac{20}{90}.$$

- Total probability:

$$P = \frac{20}{90} + \frac{20}{90} = \frac{40}{90} = \frac{4}{9}.$$

Example 41: A bag contains 4 red balls and 6 blue balls. If two balls are drawn without replacement, what is the probability that both balls are blue?

Solution:

- Probability of first ball being blue: $P(B_1) = \frac{6}{10} = 0.6$.
- Probability of second ball being blue given the first was blue: $P(B_2|B_1) = \frac{5}{9}$.
- Total probability:

$$P(B_1 \cap B_2) = P(B_1) \cdot P(B_2|B_1) = 0.6 \cdot \frac{5}{9} = \frac{1}{3}.$$

Example 42: A box contains 3 white balls and 7 black balls. If two balls are drawn without replacement, what is the probability that the second ball is white, given that the first ball is black?

Solution:

- After drawing 1 black ball, the box has 3 white and 6 black balls.
- Probability of drawing a white ball:

$$P = \frac{3}{9} = \frac{1}{3}.$$

Example 43: A deck of 52 cards is shuffled. What is the probability that the second card drawn is a queen, given that the first card drawn is a king?

Solution:

- After drawing 1 king, there are 51 cards left, including 4 queens.
- Probability of drawing a queen:

$$P = \frac{4}{51}.$$

Example 44: A bag contains 5 red balls and 5 blue balls. If two balls are drawn with replacement, what is the probability that both balls are red?

Solution:

- Probability of drawing a red ball each time: $P(R) = \frac{5}{10} = 0.5$.
- Since draws are independent, total probability:

$$P(R_1 \cap R_2) = P(R) \cdot P(R) = 0.5 \cdot 0.5 = 0.25.$$

Example 45: A box contains 2 red balls and 4 green balls. If two balls are drawn without replacement, what is the probability that the second ball is green, given that the first ball is red?

Solution:

- After drawing 1 red ball, the box has 1 red and 4 green balls.
- Probability of drawing a green ball:

$$P = \frac{4}{5}.$$

Example 46: A disease affects 3% of the population. A test for the disease is 98% accurate (true positive rate) but has a 4% false positive rate. If a person tests positive, what is the probability they have the disease?

Solution:

- Let $P(D) = 0.03$, $P(\neg D) = 0.97$.
- Let $P(T|D) = 0.98$, $P(T|\neg D) = 0.04$.
- Using Bayes' Theorem:

$$P(D|T) = \frac{P(T|D) \cdot P(D)}{P(T|D) \cdot P(D) + P(T|\neg D) \cdot P(\neg D)} = \frac{0.98 \cdot 0.03}{0.98 \cdot 0.03 + 0.04 \cdot 0.97} = 0.431.$$

Example 47: A company has two machines, A and B. Machine A produces 70% of the items, and Machine B produces 30%. The defect rate for Machine A is 2%, and for Machine B is 5%. If an item is defective, what is the probability it was produced by Machine A?

Solution:

- Let $P(A) = 0.7$, $P(B) = 0.3$.
- Let $P(D|A) = 0.02$, $P(D|B) = 0.05$.
- Using Bayes' Theorem:

$$P(A|D) = \frac{P(D|A) \cdot P(A)}{P(D|A) \cdot P(A) + P(D|B) \cdot P(B)} = \frac{0.02 \cdot 0.7}{0.02 \cdot 0.7 + 0.05 \cdot 0.3} = 0.482.$$

Example 48: A box contains 4 white balls and 3 black balls. Two balls are drawn without replacement. If the second ball is white, what is the probability that the first ball was black?

Solution:

- Let $P(B_1) = \frac{3}{7}$, $P(W_1) = \frac{4}{7}$.
- Let $P(W_2|B_1) = \frac{4}{6} = \frac{2}{3}$, $P(W_2|W_1) = \frac{3}{6} = 0.5$.
- Using Bayes' Theorem:

$$P(B_1|W_2) = \frac{P(W_2|B_1) \cdot P(B_1)}{P(W_2|B_1) \cdot P(B_1) + P(W_2|W_1) \cdot P(W_1)} = \frac{\frac{2}{3} \cdot \frac{3}{7}}{\frac{2}{3} \cdot \frac{3}{7} + 0.5 \cdot \frac{4}{7}} = 0.5.$$

Example 49: A test for a disease is 99% accurate. The disease occurs in 0.5% of the population. If a person tests positive, what is the probability they have the disease?

Solution:

- Let $P(D) = 0.005$, $P(\neg D) = 0.995$.
- Let $P(T|D) = 0.99$, $P(T|\neg D) = 0.01$.

- Using Bayes' Theorem:

$$P(D|T) = \frac{0.99 \cdot 0.005}{0.99 \cdot 0.005 + 0.01 \cdot 0.995} = 0.332.$$

Example 50: A box contains 3 red balls and 7 blue balls. Two balls are drawn without replacement. If the second ball is red, what is the probability that the first ball was blue?

Solution:

- Let $P(B_1) = \frac{7}{10} = 0.7$, $P(R_1) = \frac{3}{10} = 0.3$.
- Let $P(R_2|B_1) = \frac{3}{9} = \frac{1}{3}$, $P(R_2|R_1) = \frac{2}{9}$.
- Using Bayes' Theorem:

$$P(B_1|R_2) = \frac{P(R_2|B_1) \cdot P(B_1)}{P(R_2|B_1) \cdot P(B_1) + P(R_2|R_1) \cdot P(R_1)} = \frac{\frac{1}{3} \cdot 0.7}{\frac{1}{3} \cdot 0.7 + \frac{2}{9} \cdot 0.3} = 0.7.$$